Supplementary Material 1

Supplement to "Pre-Export Row Scaling for Generated Mixed-Integer Programs: Accuracy Contracts and Software Evaluation"

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This supplement specifies the computational evidence and its limitations. Paths are relative to the accompanying repository. Frozen results can be checked without an optimizer license. The earlier draft and intermediate runs are retained to make corrections traceable; they are not interchangeable with the final datasets.

S1 Proof for the triangular model

For $R \in \mathbb{R}^{m\times n}$, $m, n \geq 1$, set $\rho = \|R\|_2$ and

$$C_\gamma = \begin{bmatrix} I_n & 0 \\ \gamma R & \gamma I_m \end{bmatrix}, \qquad \gamma > 0.$$

The main-text proposition states that the unique optimal common coupling factor is $(1+\rho^2)^{-1/2}$ and the minimum condition number is $\sqrt{1+\rho^2}+\rho$. The interval-constrained optimum clips this factor to the interval.

Proof. An SVD of R and orthogonal transformations reduce the nontrivial spectrum to blocks $\left[\begin{smallmatrix} 1 & 0 \\ \gamma\sigma & \gamma \end{smallmatrix}\right]$, one for each singular value σ of R, with possible additional singular values 1 or γ. Each block's squared singular values have product γ^2 and sum $1+\gamma^2(1+\sigma^2)$. Their largest value increases and smallest value decreases with σ; the block at $\sigma = \rho$ therefore determines both extrema, including any additional values. If k is their singular-value ratio, then

$$k + k^{-1} = \frac{1+\gamma^2(1+\rho^2)}{\gamma} = \gamma^{-1} + \gamma(1+\rho^2).$$

The left-hand side is increasing for $k \geq 1$. The right-hand side has a unique minimum at $\gamma_* = (1+\rho^2)^{-1/2}$, decreases before that point, and increases after it. At the minimum, $k+k^{-1} = 2\sqrt{1+\rho^2}$, whose root $k \geq 1$ is $\sqrt{1+\rho^2}+\rho$. The interval result follows from the same monotonicity. For $\rho = 0$, the argument reduces to balancing the diagonal entries at one. □

This scalar result assumes already whitened local blocks and one common coupling factor. It does not solve general diagonal preconditioning and says nothing directly about simplex bases or mixed-integer search.

S2 Fixed-basis conditioning audit

Protocol. The protocol was fixed after a pilot on the first five inputs of each class (`fixed-basis-recovery-20260914-v2/`) and then applied, unchanged except for the policy set of the manuscript, to all 306 archived dispatch inputs with no exclusion: 204 Simple, 68 SG-Ter-Mer and 34 Full models. The audit build of the C++ generator writes five LP files per model through its solver-free exporter, with the following preprocessing flags.

Policy	Flags of `preprocesar`
Base	`--solodiagnostico`
Flat-GM	`--local-matricial --plano`
Flat-L2	`--local-matricial --plano --kernel-euclideo`
Role-Hybrid	`--local-matricial --acoplamiento-l2`
SA-Mat	`--local-matricial`

HiGHS 1.15.1 reads Base with small-matrix threshold 10^{-12}, drops integrality and solves the LP relaxation with the simplex method (one thread, random seed 0, 300-second limit); all 306 solves are optimal. The basis is stored by row and column names.

Each variant is read with the same reader and compared with Base before assembly: identical named rows and columns, sparsity pattern, bounds, objective, offset and integrality. A row's factor is inferred from its largest-magnitude coefficient, and the whole row and both RHS values must agree with that factor to relative tolerance 2×10^{-13}; the largest observed deviation is 2.5×10^{-16}. The basis consists of the basic structural columns of the variant's own matrix and a unit column for every basic slack.

Estimate and verification. $\|B\|_1$ is computed from the stored entries, and $\|B^{-1}\|_1$ is estimated by SciPy's `onenormest` ($t = 4$) on solves with a SuperLU factorization, with seeds 0, 1 and 2; the tables use seed 0. The seeds agree within a factor of 2.4, and 63 of 1,530 records differ by more than 1%. The largest componentwise backward error of a diagnostic solve is 1.9×10^{-8}. The script `scripts/research/verify_fixed_basis.py` imports neither the driver nor the HiGHS reader. It checks the recorded hashes, parses every LP file as text, rebuilds each basis from the stored names and compares it entry by entry with the stored matrix: all 1,530 bases match exactly. It then recomputes each estimate as $B^{-1} = Q^{-1}D$, with $Q = DB$ and D an exact power-of-two row equilibration. The two routes differ by less than 10^{-6} in 1,512 records and by at most 23% in all; since their test solutions agree to 10^{-12} in every record, the remaining differences point to the randomized estimator's iteration path rather than to the solves.

Interpretation. Row scaling changes κ_1, so the audit measures the exported representation before any solver-internal scaling. The Base estimates exceed the reciprocal of machine precision; the precision check below shows that they are nevertheless accurate on a stress sample. Every row of the 306 models has a coefficient of magnitude at least one, and $1/\|a_r\|_2 \le 1/\max_j |a_{rj}|$, so no Flat-L2 factor exceeds one. Across all rows, Flat-GM enlarges 22.5% (Simple), 32.1% (SG-Ter-Mer) and 27.3% (Full) and leaves 30.0%, 22.0% and 24.5% unscaled; Flat-L2 enlarges none and leaves 57.6%, 64.3% and 59.2% unscaled. SA-Mat and Role-Hybrid exports are byte-identical on all 306 models. SA-Mat coincides with Flat-GM on every Simple model and, in each SG-Ter-Mer and Full model, differs from it only on the 1,440 `falla_tramo` market-shortage rows, which SA-Mat and Role-Hybrid leave unscaled. The authoritative directory is `results-revision/research-audit/fixed-basis-final-20260914/`, with design, source snapshot, exports, bases, per-row factors, verification and `analysis.json`.

Table S1: Fixed-basis audit: per-instance ratios $\kappa_1(\text{policy})/\kappa_1(\text{Base})$ on the common basis. "Improved" counts ratios below one.

Class	Policy	n	Median ratio	Minimum	Maximum	Improved
Simple	Flat-GM	204	2.4×10^{-7}	1.9×10^{-8}	1.7×10^{-3}	204
Simple	Flat-L2	204	1.57	3.3×10^{-4}	2.7×10^{2}	32
Simple	Role-Hybrid	204	2.4×10^{-7}	1.9×10^{-8}	1.7×10^{-3}	204
Simple	SA-Mat	204	2.4×10^{-7}	1.9×10^{-8}	1.7×10^{-3}	204
SG-Ter-Mer	Flat-GM	68	7.3×10^{-9}	1.5×10^{-9}	1.2×10^{-2}	68
SG-Ter-Mer	Flat-L2	68	1.51	8.1×10^{-4}	18.61	10
SG-Ter-Mer	Role-Hybrid	68	7.3×10^{-9}	1.5×10^{-9}	0.57	68
SG-Ter-Mer	SA-Mat	68	7.3×10^{-9}	1.5×10^{-9}	0.57	68
Full	Flat-GM	34	3.5×10^{-7}	9.3×10^{-9}	3.1×10^{-6}	34
Full	Flat-L2	34	1.49	0.85	1.5×10^{2}	2
Full	Role-Hybrid	34	3.5×10^{-7}	9.3×10^{-9}	3.1×10^{-6}	34
Full	SA-Mat	34	3.5×10^{-7}	9.3×10^{-9}	3.1×10^{-6}	34

Precision check. Because the unscaled values exceed the reciprocal of machine precision, the script `scripts/research/verify_fixed_basis_precision.py` checks their accuracy on a stress sample fixed before computation: in each class, the model with the largest Base estimate, the model with the largest Flat-GM/Base estimated ratio, and one other model drawn with seed 20260915. In Full the first two coincide (instance014), which gives eight models and 24 bases under Base, Flat-GM and Flat-L2. For each hash-checked basis, $\|B\|_1$ is computed in exact rational arithmetic. Every column of B^{-1} is solved in double precision; with x_j the computed column and ρ_j a bound on $\|Bx_j - e_j\|_1$ that includes the rounding-error terms of the residual evaluation, $\|B^{-1}\|_1 \le \max_j \|x_j\|_1/(1 - \max_j \rho_j)$ whenever $\max_j \rho_j < 1$. This upper endpoint is evaluated in floating point with explicit error terms rather than directed rounding; the implementation does not provide a formal rounding certificate for that endpoint. We therefore call the resulting ranges diagnostic brackets, not certified enclosures. For the five largest columns, the solution is refined with residuals computed exactly in rational arithmetic on the stored binary64 data; since $\|B^{-1}\|_1 \ge \|x\|_1/\|Bx\|_1$ for every nonzero x, the exact quotient certifies a lower bound.

The largest residual bound is 6.0×10^{-3}, and refinement stops within two steps. The diagnostic brackets have relative width at most 6.0×10^{-3} for Base, 4.8×10^{-3} for Flat-L2 and 2.0×10^{-5} for Flat-GM, and the computed upper estimate of the Flat-GM/Base ratio agrees with the estimated ratio to 2.1×10^{-5} in all eight models (Table S2). Twenty-three of the 24 estimates lie inside their brackets. The Flat-L2 estimate for Full instance046 is 2.4 times smaller than its certified lower bound, as a lower-bound estimator can be; this makes Flat-L2 less favorable, not more. Outputs are in `precision-check/` within the audit directory.

Archived fixed-incumbent computation. An earlier computation (`results-revision/fixed-basis/`) used a different reference basis: Gurobi solved each Base MIP (relative gap 10^{-3}, 16 threads), fixed the integers at its incumbent and took the basis of the resulting LP. Its archived medians point in the same direction with smaller magnitudes; for SA-Mat relative to Base they are 1.2×10^{-3} (Simple), 3.6×10^{-4} (Full) and 0.41 (SG-Ter-Mer). The magnitudes therefore depend on the reference basis. That computation did not retain its exports, basis identities or binary hash, and its multithreaded incumbents cannot be replayed, so it is not used as evidence. The provenance audit is in `results-revision/research-audit/fixed-basis-recovery-20260914-v2/`.

Table S2: Precision check on the stress sample. "Base $\kappa_1 \geq$" is the certified lower bound. "Bracket width" is the largest relative width (computed upper endpoint over lower bound, minus one) among Base, Flat-GM and Flat-L2. The last two columns give the estimated ratio κ_1(Flat-GM)/κ_1(Base) and its upper estimate, the floating-point upper endpoint for Flat-GM divided by the certified lower bound for Base. Only the lower bounds carry the rational certification described above.

Class	Model	Base $\kappa_1 \geq$	Bracket width	Flat-GM/Base estimate	Upper estimate
Simple	caso32e	9.3×10^{19}	1.5×10^{-4}	2.405×10^{-6}	2.405×10^{-6}
Simple	caso05f	1.0×10^{19}	3.7×10^{-3}	1.681×10^{-3}	1.681×10^{-3}
Simple	caso27f	8.7×10^{18}	1.9×10^{-5}	1.077×10^{-7}	1.077×10^{-7}
SG-Ter-Mer	instance067	1.2×10^{21}	4.1×10^{-3}	6.247×10^{-6}	6.247×10^{-6}
SG-Ter-Mer	instance028	4.0×10^{11}	1.5×10^{-6}	1.216×10^{-2}	1.216×10^{-2}
SG-Ter-Mer	instance074	4.0×10^{20}	9.5×10^{-5}	2.277×10^{-9}	2.277×10^{-9}
Full	instance014	1.2×10^{22}	6.0×10^{-3}	3.055×10^{-6}	3.055×10^{-6}
Full	instance046	4.7×10^{20}	2.7×10^{-4}	6.046×10^{-7}	6.046×10^{-7}

Table S3: Computing environment by campaign. Missing historical machine identities are not reconstructed. Local wall times depend on the shared-machine load; offline reference LP settings are stated in the corresponding protocols.

Campaign	Optimizer	Thr.	Limit	Machine	Reported quantity
Historical experiments					
Fixed-basis audit	HiGHS 1.15.1 (LP only)	1	300 s	not recorded	κ_1
Matrix audit	none	—	—	not recorded	κ_2^+, identity
Operational campaign	Gurobi 13.0.1	16	1,800 s	Xeon Gold 6138, Debian 12	PAR10 (s), work units
Replication, seed 7	Gurobi 13.0.1	16	1,800 s	ClusterUY, node not recorded	work units
Replication, CPLEX	CPLEX 22.1.2.0	16	1,800 s	ClusterUY, node not recorded	deterministic ticks
Stress experiment	HiGHS 1.15.1, Gurobi 13.0.2, CPLEX 22.1.2.0	1	10 s	not recorded	verified counts
Local OMS campaigns: Intel i7-13620H, Ubuntu 22.04.5 LTS					
Ruiz/stage comparison	Gurobi 13.0.2	4	120 s	shared local machine	Work, PAR10 (s)
Activation follow-up	Gurobi 13.0.2	4	120 s	shared local machine	budgets, Work
Common-incumbent control	Gurobi 13.0.2	4	12 s / 12 Work	shared local machine	Work, counts
Matched IF control	Gurobi 13.0.2	4	120 s	shared local machine	Work, total (s)
Matched accuracy/remedies	Gurobi 13.0.2	4	120 s	shared local; reboot retained	Work, total (s)

S3 Computing environment

Table S3 separates the historical experiments from the local OMS campaigns. The historical operational PAR10, local Ruiz comparison and full-process measurements in Tables S15–S16 use wall seconds. These depend on hardware and load; the local timings describe the measured prototype on a shared machine. The 144-call comparison also spans a documented reboot. Native Work and deterministic ticks are reported within their respective solver/version, without equating them with wall time or with each other.

All 240 historical operational logs identify the same processor and operating system. The laptop logs in `operational/precision-smoke/` belong to a separate test. The two historical replications used ClusterUY/SLURM with 16 cores per task; their node models were not recorded. Historical fixed-basis, matrix and stress machine identities remain unrecorded. The newer dispatch logs identify an Intel i7-13620H running Ubuntu 22.04.5 LTS. Original records, not the current host, supply these identities.

S4 Operational evidence and reconstruction

The final historical campaign assigned 15 instances in each of two dispatch families, four policies, and two relative MIP gaps. Thus there are 240 assigned Gurobi 13.0.1 runs (Table S3). All use seed 42, 16 threads and a 1,800-second limit. The four policies are Base, Flat-GM, Flat-L2 and Role-Hybrid. The assigned instances are

> Simple: `caso01a`, `caso03b`, `caso05d`, `caso10e`, `caso14a`, `caso20c`, `caso22d`, `caso24f`, `caso27a`, `caso29c`, `caso31e`, `caso33f`, `caso42b`, `caso44c`, `caso46e`.
> SG-Ter-Mer: `instance001`, `instance005`, `instance014`, `instance022`, `instance027`, `instance031`, `instance042`, `instance046`, `instance051`, `instance058`, `instance068`, `instance072`, `instance077`, `instance081`, `instance092`.

They are those of the historical campaign and are spread across the ordering of each class, but the rule that produced them was not recorded, so this is a description of the subsets rather than evidence of a pre-registered design. What can be checked is that they predate the analysis: the runs are reconstructed from logs retrieved from the cluster, and the frozen tables of the two replications reuse the same two lists. The analysis retains the assigned subsets and includes every non-completion in the primary runtime endpoint.

For assigned instance i in a cell, let t_i be the recorded runtime and define $T_i = t_i$ for completion and $T_i = 18{,}000$ otherwise. The primary summary is $15^{-1} \sum_i T_i$. This PAR10 endpoint measures operational completion under the specified stopping criteria. It is not an exact primal or optimality certificate. In particular, a solver-completed point with a concerning original residual is still counted as completed, with feasibility diagnostics reported separately.

Scenario defect in the SG-Ter-Mer instances. The market dispatch of this family reads a local demand series and a total demand series. In the historical instances the total series was rebuilt from a noisy rescaling of residual demand, and in 39 of the 68 instances it falls below the local demand in at least one hour: 9 of the 15 instances assigned above, among them instance077, where the inversion occurs in 22 hours and reaches 259 units. The exported models remain well-posed MILPs and every comparison in this paper is between representations of the same model, so no reported equivalence, residual or effort measurement is affected. What the defect does mean is that those instances are not faithful market scenarios and that the physical reading of their solutions is unreliable. A corrected version of the family, rebuilt from the authoritative demand series, exists outside this work and shows no inversion; no result reported here uses it. The counts above are produced by `scripts/research/check_sgtm_demand.py` from the instance files in the archive.

Does the defect drive the SG-Ter-Mer result? Six of the fifteen instances are free of the inversion. Table S4 repeats the two endpoints on that subset, with the same work measure and the same common-completion pool as the main text. The conditional gain does not depend on the defective scenarios: the Flat-GM work ratio is 0.452 at a 1% gap and 0.502 at 0.1% on the six, against 0.698 and 0.682 on all fifteen. The failure-sensitive reversal does depend on them, for a reason worth stating: the instance that times out under every scaled policy, instance077, is one of the nine, so on the defect-free subset every scaled policy completes and PAR10 follows the conditional gain instead of contradicting it. The reversal reported in the main text therefore rests on a single instance, which is what makes it a statement about failure sensitivity rather than about typical effort.

Table S4: SG-Ter-Mer reconstruction on all fifteen assigned instances and on the six without the demand inversion. Work ratios are geometric means against Base on the instances every policy completed; PAR10 is the arithmetic mean over the assigned pool, with 18,000 seconds for a non-completion.

		Work ratio vs Base		Mean PAR10 (s)	
Gap	Policy	All 15	Defect-free 6	All 15	Defect-free 6
1%	Base	—	—	116.2	22.8
	Flat-GM	0.698	0.452	1216.5	5.9
	Flat-L2	0.863	0.844	1217.6	14.4
	Role-Hybrid	0.745	0.470	1217.3	5.4
0.1%	Base	—	—	134.1	25.0
	Flat-GM	0.682	0.502	1216.2	7.1
	Flat-L2	0.802	0.838	1216.0	13.6
	Role-Hybrid	0.726	0.493	1216.9	6.2

Table S5: Reconstructed operational campaign. Arithmetic mean PAR10 in seconds on the 15 assigned instances per cell; a timeout contributes 18,000 seconds. Parentheses give completed runs. All rows concern Gurobi.

Class	Gap	Base	Flat-GM	Flat-L2	Role-Hybrid
Simple	1%	11.90 (15)	2.12 (15)	20.44 (15)	2.11 (15)
Simple	0.1%	52.00 (15)	72.98 (15)	33.17 (15)	72.63 (15)
SG-Ter-Mer	1%	116.15 (15)	1216.51 (14)	1217.60 (14)	1217.34 (14)
SG-Ter-Mer	0.1%	134.08 (15)	1216.24 (14)	1216.05 (14)	1216.91 (14)

Work and uncertainty. The old runner's scaled-policy Work markers inherited a fixed, zero-decimal stream format. Native Gurobi summaries instead retain two decimals. The reconstruction uses those native summaries for every policy, at their common 0.01-unit resolution. It never combines a Work value with a timeout penalty measured in seconds. Byte equality of rounded Work or node counts cannot prove equality of exported models.

Secondary Work ratios use only the instances completed by every policy in a given family/gap cell. Ratios are paired by instance; their geometric mean is $\exp(n^{-1}\sum_i \log(W_{i,p}/W_{i,\mathrm{Base}}))$. Percentile intervals use 10,000 paired bootstrap samples, seed 20260908. These are exploratory intervals, without multiplicity adjustment, and are conditional on the common completion pool. A separate sensitivity calculation uses the intervals of width 0.01 implied by rounding each native Work value. Neither calculation accounts for the excluded timeout's unobserved completion effort.

Flat-GM/Base on the 14 common completed SG-Ter-Mer instances has ratios 0.698 with interval $[0.486, 0.944]$ at 1% gap, and 0.682 with $[0.470, 0.925]$ at 0.1%. However, instance077 completes under Base and times out under all three scaled policies at both gaps. For Simple at 1%, Flat-L2/Base has Work ratio 1.792 with interval $[1.033, 3.542]$. These conditional observations do not establish a uniform solver advantage. Table S5 gives the complete PAR10 summary.

Replication under a second seed and a second solver. The reconstruction fixes the solver, the seed and the thread count. To separate what is stable from what belongs to that configuration, the same 30 instances, four policies, two gaps, 1,800-second limit and 16 threads were rerun twice, changing one factor at a time: Gurobi with seed 7, and CPLEX with the original seed 42. The run tables were frozen before the runs, the policy order within an instance is seeded by the instance

Table S6: Paired effort ratio against Base on the instances completed by every policy, with bootstrap intervals, for the reconstruction and its two replications. Values below one favour the scaled policy.

	Gurobi, seed 42	Gurobi, seed 7	CPLEX, seed 42
Simple, gap 1%			
Flat-GM	0.631 [0.31, 1.03]	0.783 [0.47, 1.03]	0.609 [0.28, 1.04]
Flat-L2	1.792 [1.03, 3.54]	2.226 [1.16, 4.70]	1.306 [0.39, 4.26]
Role-Hybrid	0.631 [0.31, 1.03]	0.783 [0.47, 1.04]	0.609 [0.28, 1.04]
Simple, gap 0.1%			
Flat-GM	1.063 [0.89, 1.25]	1.002 [0.91, 1.10]	0.837 [0.53, 1.15]
Flat-L2	1.385 [0.90, 2.33]	1.524 [1.05, 2.41]	1.425 [0.52, 3.64]
Role-Hybrid	1.063 [0.89, 1.24]	1.002 [0.91, 1.10]	0.837 [0.53, 1.14]
SG-Ter-Mer, gap 1%			
Flat-GM	0.698 [0.49, 0.94]	0.778 [0.53, 1.06]	1.078 [0.99, 1.18]
Flat-L2	0.863 [0.72, 1.03]	1.115 [0.92, 1.38]	0.955 [0.92, 0.99]
Role-Hybrid	0.744 [0.49, 1.04]	0.775 [0.53, 1.06]	1.053 [0.98, 1.15]
SG-Ter-Mer, gap 0.1%			
Flat-GM	0.682 [0.47, 0.92]	0.777 [0.53, 1.06]	1.077 [0.99, 1.18]
Flat-L2	0.802 [0.65, 0.98]	1.110 [0.93, 1.36]	0.994 [0.93, 1.09]
Role-Hybrid	0.726 [0.48, 1.00]	0.779 [0.53, 1.06]	1.052 [0.97, 1.16]

digest, and the records carry the same verification as the campaign. Table S6 reports the paired effort ratio against Base on the instances every policy completed. Effort is each solver's own measure, work units for Gurobi and deterministic ticks for CPLEX, so a column is read within its solver and never across. Runs classified as infrastructure failures were repeated and solver timeouts were not; there were two such episodes, the eight Gurobi runs of one instance that failed while the licence server was saturated, and the eight CPLEX runs of instance077, repeated with more memory after one of them was killed by the cgroup limit. The instance that reverses the PAR10 ordering in the reconstruction, instance077, behaves differently in every condition: under Gurobi at seed 7 all eight of its runs reach the limit, Base included, while under CPLEX the run that reaches the limit is Base at the 1% gap and Role-Hybrid at 0.1%.

Original residual normalization. For an equality, the absolute residual is $|a_r^T x - b_r|$. The RHS-normalized residual divides by $1 + |b_r|$, whereas the activity-normalized residual divides by $\max\{1, |b_r|, \sum_j |a_{rj} x_j|\}$. For inequalities the positive violation replaces the absolute equality residual. Neither is a supplied engineering acceptance budget. For caso46e, Flat-L2 at 1% produces an absolute equality residual about 4.97 on a row with zero RHS. Its RHS-normalized value is still 4.97, while the run's maximum activity-normalized residual is 1.15×10^{-7}. Calling the former a 497% physical error would be unjustified.

The directory `results-revision/research-audit/operational/` contains native cluster logs, the reconstructed run table, cell summaries, paired contrasts, original input hashes and analysis settings. The script `scripts/audit_operational_evidence.py` generates these summaries and the main-text table. A separate two-run smoke check under `precision-smoke/` verifies that the corrected C++ Work marker preserves the solver API's precision; it does not replace any of the 240 historical observations.

Historical stage ablation. The earlier preprocessing campaign included a stage ablation under Gurobi. It predates the attribution analysis of the present article and is reported here only as exploratory evidence about solver effort. The structured rules had three conceptual stages: local-row normalization, block-scale construction and coupling-row scaling. The ablation retained either the local stage alone or the coupling stage alone and compared them with Base and the full rule on the same instance banks, at relative MIP gaps of 1% and 0.1%.

Table S7 gives the structured rule on all three classes. Local-only normalization reproduces the full rule's shifted geometric mean of solver time in every cell, to within 0.5% except on Simple at a 1% gap, where the two differ by 0.01 s. Coupling-only scaling never improves on Base: it matches it on Simple and SG-Ter-Mer and is 1 to 2% above it on Full. The matrix rule, recorded in the same files, agrees cell by cell with one exception worth stating: on Full at a 0.1% gap it does not improve on Base at all (396.71 s for the full rule against 373.58 s), and local-only tracks it there as elsewhere (394.67 s). The attribution is therefore about where a stage-wise difference sits, not a claim that either rule always helps.

This does not contradict the coupling-kernel attribution of the article. That concerns which kernel explains a representation-level conditioning difference under matched controls; this ablation concerns where an operational Gurobi time difference arose in an earlier campaign. The endpoints and the comparisons are different.

Table S7: Historical stage ablation, structured rule. Shifted geometric mean of solver time in seconds, Gurobi. The ablation belongs to the earlier preprocessing campaign and is not a rerun of the present policy set; `make research-stage-ablation` recomputes it from the stored runs in `results-revision/r5-ablation/`.

Class	Gap	n	Base	Full rule	Local only	Coupling only
Simple	1%	204	3.73	2.02	2.01	3.74
Simple	0.1%	204	9.61	6.41	6.39	9.59
SG-Ter-Mer	1%	68	14.42	8.55	8.58	14.45
SG-Ter-Mer	0.1%	68	16.15	9.79	9.76	16.16
Full	1%	34	82.55	71.41	70.94	83.69
Full	0.1%	34	373.58	325.73	325.91	376.29

S5 Fresh matrix audit

Ten seeds, 20260909–20260918, are crossed with exponent ranges $[-R, R]$ for $R \in \{1, 2, 3\}$, two disturbance severities (3 and 6), and three coupling patterns: `coupling`, `coupling_uniform`, and `mixed`. There are 180 instances and six exports per instance: Base, SA-Mat, Flat-GM, Flat-L2, Role-Hybrid and SA-Mat-permB. A solver-free DummyLp backend writes the 1,080 LP files.

The parser compares named rows and columns, coefficient supports, row senses, right-hand sides, and the remaining objective/domain sections. It infers each row's positive factor from a nonzero pivot coefficient and checks the entire row and RHS at relative tolerance 10^{-13}. All final exports pass. This remains a floating-point proportionality check; it is not a rational proof of exact scaling. Byte hashes supply the stronger literal-identity claims where they hold.

The numerical reference rank is determined from row-L2-normalized Base using a relative threshold 10^{-12}. Every variant then uses that same rank for the spectral diagnostic. Independent rank thresholds on each export disagree on 25 of 1,080 cases. Very large unscaled condition estimates near binary64 resolution are therefore not interpreted quantitatively. The paired scaled diagnostics

are below 6.5×10^6.

The authoritative directory is `results-revision/research-audit/identifiability-final/`. Its manifest records the generator binary and script hashes, settings, named comparisons and all exports. The retained `identifiability-confirmatory/` directory used the former fixed precision serializer and is superseded for current matrix claims.

S6 Frozen binary stress protocol

The experiment crosses three binary families, eight seeds (100–107), and four coefficient exponents (0, 3, 6, 9): 96 models with 12 binary variables each. Each model is run under two presolve settings, three solvers, and nine policies, giving 5,184 planned configurations. Budget-round abstains on 432 configurations before calling an optimizer; hence there are 4,752 actual calls. These configurations are not statistically independent observations.

For seed s, NumPy's default generator is seeded by $20260908 + s$. The cardinality family has one all-ones row. Packing uses three rows of uniformly sampled integers 1 through 9. Signed rows use integers from -9 through 9, replacing zeros by one. For row r, $k_r = \max\{2, \lfloor 0.55 \sum_j \max(W_{rj}, 0) \rfloor\}$; profits are uniform integers 1 through 30. The represented constraints are $10^e W x \leq 10^e k - 0.1\mathbf{1}$. Integer enumeration proves the binary optimum for each stored model. Changing e also changes the relaxation; it is not pure row re-emission. The budget is 0.01 in the declared canonical stress-test units.

All three solvers use one thread, the instance seed, a ten-second limit and zero target absolute and relative MIP gaps. Primal and integrality tolerances are 10^{-6}, except GM-tight, which uses 10^{-9}. Gurobi's enabled-presolve arm uses its automatic presolve setting; HiGHS uses presolve on. Internal solver scaling remains at its default. Recorded versions are Python 3.10.12, NumPy 2.2.6, HiGHS 1.15.1 and Gurobi 13.0.2. Solving uses the Python APIs rather than text-file import.

Base, GM and L2 scale all rows according to their stated kernels, without a skip band. Budget-only uses ϵ/δ_r. Budget-GM projects GM onto the joint budget/coefficient interval with window $[10^{-9}, 10^9]$. Budget-round additionally reserves $10^{-6} \sum_j |a_{rj}|$ for rounding and rejects the whole model if any row's contract is incompatible. Dyadic-GM imposes the coefficient window and exact binary64 multiplication, including the RHS; Budget-dyadic also imposes the residual budget. The latter two policies were added after the initial floating-endpoint audit and are exploratory, not a new holdout experiment.

The sequence of preserved datasets under `results-revision/research-audit/` is:

Directory	Interpretation
`budget-pilot`	Single exploratory seed 0; six initial policies.
`budget-confirmatory`	Seeds 100–107; seven policies; original nearest-rounded endpoint.
`budget-confirmatory-v2`	Same seven policies and seeds; exact endpoints rounded inward.
`budget-final`	Authoritative nine-policy run; inward endpoints and two exploratory dyadic controls.

An attractive result from an intermediate implementation is not substituted for the corrected final result. In particular, the earlier HiGHS Budget-GM count of 191 verified configurations becomes 182 with correctly inward-rounded endpoints.

Paired outcomes and factor distances. The paired comparison in the main text uses identical model, presolve and solver keys; its counts are regenerated by `scripts/research/analyze_budget_experiment.py`, which also writes `budget-final/paired-discordance.json`. Budget-only and

Table S8: Enumerated stress experiment, 192 planned configurations per policy and solver. "Verified" means that the returned point passes the declared acceptance rule and its rounded assignment attains the independently enumerated optimum. "Other" includes unverified outputs and solver errors; "Abstain" means no optimizer call was attempted. Abstention is not a performance success. The separated GCD-floor row is the later integer-aware control, with the same denominator and acceptance rule.

	HiGHS 1.15.1			Gurobi 13.0.2			CPLEX 22.1.2		
Method	Verified	Other	Abstain	Verified	Other	Abstain	Verified	Other	Abstain
Base	167	25	0	156	36	0	150	42	0
GM	141	51	0	140	52	0	140	52	0
L2	141	51	0	139	53	0	140	52	0
GM-tight	166	26	0	166	26	0	166	26	0
Dyadic-GM	142	50	0	140	52	0	140	52	0
Budget-only	191	1	0	151	41	0	140	52	0
Budget-GM	182	10	0	152	40	0	140	52	0
Budget-dyadic	161	31	0	153	39	0	140	52	0
Budget-round	48	0	144	48	0	144	48	0	144
GCD-floor	192	0	0	192	0	0	192	0	0

Budget-GM select different factors at multipliers 10^0 and 10^3, where both verify every configuration on all three solvers. At 10^6 and 10^9 the budget endpoint binds, and Budget-GM returns the inward-rounded endpoint, at most one unit in the last place from Budget-only's factor ϵ/δ on every row of every model. All ten configurations whose verification outcomes differ between the two policies lie at these multipliers. The failure categories quoted in the main text are taken from `results-revision/research-audit/exploration-failures/summary.json` for HiGHS and Gurobi and `exploration-failures-cplex/summary.json` for CPLEX, both written by the same script, which classifies each returned point with an infeasible rounded assignment by whether it violates the original row budget and whether it violates the exported tolerance.

Classical integer-row controls. The follow-up in `results-revision/research-audit/exploration-lattice/` uses the same 96 stored models, three solvers and two presolve settings. For an integral row $a^T z \le b$, let $g = \gcd_j |a_j|$ and replace it by $(a/g)^T z \le \lfloor b/g \rfloor$. Exact rational division of the stored RHS determines the floor. GCD-floor exports this row directly; GCD-floor-GM adds floating GM normalization; GCD-midgap-GM first replaces the floored RHS by $\lfloor b/g \rfloor + 1/2$ and then applies GM. The three controls account for 1,728 further calls, all yielding verified original-model optima. The midpoint operation is integer-equivalent before multiplication and weakens the LP relative to flooring. These are established reductions or arithmetic controls, not new scaling proposals.

The direct GCD-floor export preserves all 96 binary feasible sets exactly and verifies 192/192 configurations per solver. Independent enumeration of the actual binary64 exports also finds equivalence for all 96 GCD-midgap-GM models, but detects excluded feasible boundary points in 60 of the GCD-floor-GM models. Thus successful returned optima do not establish equivalence of every represented constraint. The main comparison foregrounds GCD-floor, whose transformation and export are both exact for these inputs. This arithmetic distinction also explains why an exact-export condition is separate from selecting a mathematically admissible real factor.

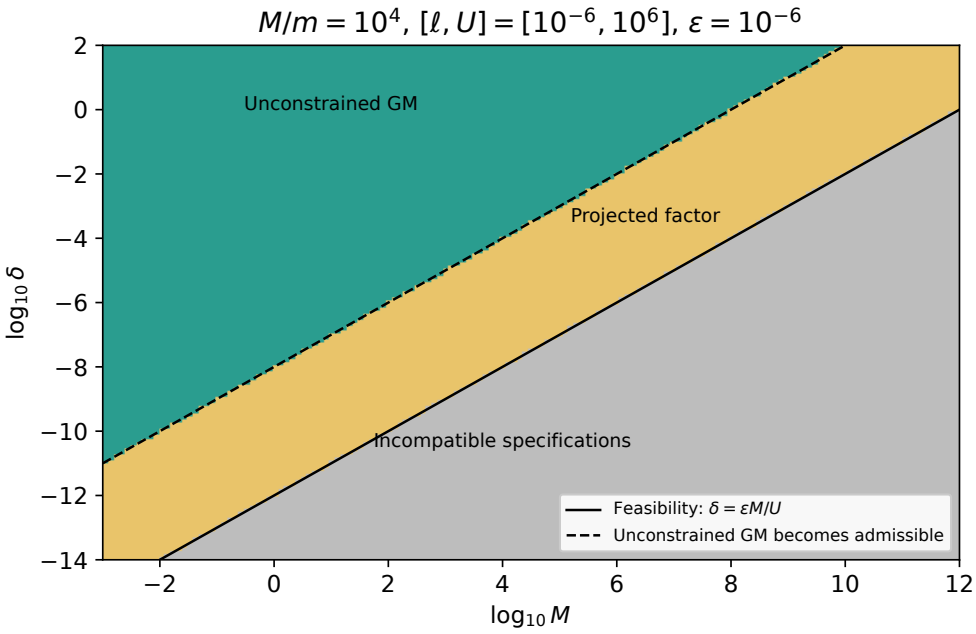


Figure S1: Exact compatibility regions for a row with fixed within-row spread. Below the solid line, the residual budget conflicts with the coefficient ceiling. In the middle region the constrained minimax factor is projected onto the admissible interval; above the dashed line ordinary GM already satisfies it. This diagram concerns the conditional row contract, before reserving any integer-rounding allowance

S7 Historical single-instance budget sensitivity

The historical caso46e illustration uses budgets declared by a modeller, not measured from an operating tolerance, so the natural objection is that admissibility was read off a convenient choice. Table S9 scales all six declared budgets by a common multiplier and reports, per constraint group, the tightest multiplier at which each kernel is still admissible on every row of the group. Admissibility is monotone in the budget — a larger δ only widens the interval of Proposition 5.1 — so the tightest multiplier with full coverage is the point at which the contract starts to bind, and the data confirm the monotonicity on every group and kernel.

All three kernels remain admissible on all 4,320 rows down to a multiplier of 10^{-4}: the declared budgets could be tightened ten thousandfold before any of them loses a row. The two kernels the main text compares, the unit factor of Base and the geometric mean of Flat-GM, survive to 10^{-5}. Below that the groups separate in the way the proposition predicts: the water balances, whose coefficients span the widest range, keep admitting the unit factor down to 10^{-9}, while the Euclidean kernel of Flat-L2 is the first to leave the admissible interval on the four groups written in megawatts. The conclusion of the main text is therefore not an artifact of the particular numbers declared; it would take a tolerance four orders of magnitude tighter than a tenth of a megawatt to make the choice of kernel a question of admissibility rather than of conditioning.

The analysis is reproducible without an optimizer. The exported LP it reads is stored beside the results, `results-revision/research-audit/application-budgets/caso46e-dummylp.lp.gz`, and `make research-application-export` re-exports it from the instance with the solver-free build and compares the two byte by byte; `make research-application-budgets` regenerates both tables

Table S9: Tightening the declared budgets of the application table. For each constraint group of caso46e, the smallest common multiplier of the declared budgets at which the kernel is still admissible on every row of the group, with $\epsilon = 10^{-6}$ and the coefficient window $[10^{-9}, 10^{9}]$. A multiplier of one is the budget declared in the main text; smaller means a tighter tolerance. The sweep covers 10^{-16} to 10^{2}.

Constraint group	Rows	Tightest admissible multiplier		
		Base	Flat-GM	Flat-L2
Generation identities, demand cap	1,080	10^{-5}	10^{-5}	10^{-4}
Generation definition, Bonete/Palmar	720	10^{-5}	10^{-5}	10^{-4}
Production envelope tangents	1,080	10^{-5}	10^{-5}	10^{-4}
Water balance, Bonete/Palmar	720	10^{-9}	10^{-7}	10^{-5}
Water balance, Baygorria	360	10^{-9}	10^{-5}	10^{-5}
Generation definition, Baygorria	360	10^{-12}	10^{-9}	10^{-4}
All groups	4,320	10^{-5}	10^{-5}	10^{-4}

from it. That export is written by the repository's own serializer rather than by Gurobi, as the archived one was: the two differ in the last digits of their decimal output, which moves the reported quantities of one group by a relative 10^{-15} and nothing in either table.

S8 Additional OMS dispatch validation

The additional protocol is frozen in `results-revision/research-audit/oms-close-20261013/design.json`. It uses the same 30 operational inputs, Gurobi 13.0.2, four threads, seed 42, 120 seconds and a 1% gap. The separate pilot does not enter the results. The six-model stage ablation selects positions 1, 8 and 15 in each archived class list. All 132 exports retain objective, bounds and domains and are checked by named row proportionality; the partial-stage factors have disjoint supports and multiply to the full Role-Hybrid factors. A mechanical LP-serializer correction, made before the first new optimization call, is recorded in `protocol-amendments.json`.

Execution resumed after an interruption with identical hardware and settings. Four observations whose incumbent archives were empty were repeated under the unchanged protocol and replaced, with the superseded CSVs and recovery criterion preserved in `artifact-recovery.json`. They are not extra observations in the paired sample.

Ruiz-R4 freezes column factors at one and performs four coefficient-only infinity-norm half-steps. Its final cumulative factor passes the same identity band and coefficient window as the other policies. The historical C++ Ruiz option is not used. All compressed LPs, named reference bases, factors, solution vectors and logs are stored. `comparison-summary.csv` and `ablation-summary.csv` retain the paired sample size for each aggregate; `runs.csv` preserves unfinished runs and their PAR10 penalties. The 120-second local experiment is never pooled with the historical 1,800-second operational campaign.

The three physical precision scenarios use 0.01, 0.1 and 1 MW for power, and 100, 1,000 and 10,000 cubic metres for water. They are declared study objectives, not operator tolerances. Algebraic budgets are read from the named generation/storage coefficient or the hourly-flow conversion. Every row-level proposed factor, interval and applied factor is retained in `factor-contracts.csv.gz`; it includes the partial-stage policies on their six-model subset. The exported-row hypothesis is 10^{-6} with zero rounding allowance for the interval audit. Actual physical residuals and pointwise coefficient/RHS export defects are computed separately by exact rational arithmetic on binary64

generator data and saved incumbents. `physical-verification.csv` records these results by physical family and counts observed violations of the exported-row hypothesis. Whole-model row, bound and integrality diagnostics remain floating-point; dispatch optimality is not independently certified. `make research-oms-check` replays the preserved exports and points, checks every interval with rational endpoints and recomputes the comparison aggregates, without calling an optimizer. The main text reports the compact comparison, final-policy ablation and application-level budget sensitivity.

S9 Cross-run certificate audit

This exploratory audit, referenced briefly in the article's discussion, compares reported objectives and dual bounds with numerically screened points. It includes an objective difference of 54.7% in the example below, without exact dispatch feasibility certification or causal attribution. The separate activation study in Section S11 supplies independently checked exact feasible witnesses of the actual imported exports.

The audit uses two stored campaigns: the solver-robustness campaign, which was prepared, partly run and then stopped by decision, and the two replications of the operational reconstruction. Neither is a census and neither was designed to estimate how often a solver contradicts itself, so the counts below describe what occurred in the runs that exist.

The comparison rests on the stored verification. Each run's returned point is re-evaluated against the original unscaled model, so a point accepted on a scaled export is still checked where it has to hold, and `verif_solver_obj_original` is its objective there. A point the campaign tolerance merely accepted can be slightly infeasible, and an infeasible point bounds nothing, so both the article and Table S10 use only the *strict* tier: witnesses whose relative activity residual, bound violation and integrality violation against the original model are each at most 10^{-9}. This is a numerical screen, not exact feasibility certification or a bound on objective perturbation. The stopped campaign contains 4,542 recorded runs: 76 reported bounds lie above a strictly screened witness across 26 instances, and 25 of those runs across nine instances also report completion beyond the requested gap. In the 480-run operational replication, the corresponding counts are 20 across six instances and nine across four. Base accounts for five of the 25 completion discrepancies in the stopped campaign and one of nine in the replication. These are counts of stored diagnostic discrepancies, not population rates or independently certified invalid solver bounds. The looser tier changes one run and one instance, in the weaker of the two levels of the stopped campaign (77 runs across 27 instances instead of 76 across 26) and nothing at all elsewhere; the stored CSVs carry both tiers in their own columns.

Table S10: Cross-run certificate audit by level and policy. "Dual bound" counts runs whose reported bound lies above a numerically screened point of the same model; "Certificate" counts the subset that also reported completion further from that point than the requested gap. Runs and instances are the totals of that level.

Campaign	Level	Runs	Inst.	Base	Flat-GM	Flat-L2	Role-Hybrid
Stopped campaign	Dual bound	76	26	6	28	12	30
	Certificate	25	9	5	9	1	10
Operational replication	Dual bound	20	6	1	6	3	10
	Certificate	9	4	1	3	1	4

The extreme case. On `v1_simple_000293`, one of the recombined Simple models, eight runs exist at a 1% gap and seed 1: four policies under each of Gurobi 13.0.1 and CPLEX 22.1.2, all recorded against the same `instance_sha256`. Seven of them return an objective between -2.893×10^8 and -2.913×10^8. The eighth, Gurobi on the unscaled Base export, returns -1.317×10^8 with status optimal, a reported gap of zero and a dual bound equal to its objective. Its stored record shows one node explored against a root gap of 1.22, a wall time of 6 s, and three solver warnings: large bounds, large matrix coefficients and large right-hand sides. Its returned point also passes the original-model numerical screen, with relative activity violation 3.2×10^{-14}; the discrepancy concerns objectives and bounds, without certifying exact feasibility or localizing a cause. The strict witness of this instance is Gurobi on the Flat-GM export at -2.909×10^8, with relative activity residual 1.1×10^{-13} and no bound or integrality violation; the reported objective exceeds it by 1.592×10^8, which is 54.7% of the magnitude of the witness. A looser witness, Gurobi on Flat-L2 at -2.913×10^8, does not meet the strict residual threshold and is not used. CPLEX on the same unscaled file explored 6,151 nodes, emitted no warnings and reached -2.904×10^8.

The identical instance hashes support comparison of the stored outcomes. The discrepant completion occurs on the unscaled export too, so the audit cannot attribute it uniquely to external scaling. Neither exact dispatch optimality nor a cause inside the solver is established by these records. The warnings motivate further numerical diagnosis; they do not prove a particular failure mechanism.

S10 Independent verification and arithmetic checks

The independent script `scripts/research/verify_saved_experiment.py` imports neither the experiment runner nor an optimizer. For each of the 96 stored models it enumerates all 4,096 binary assignments using integer activities, checks the reported optimum, and then recomputes the acceptance decision from the saved point. It checks the HiGHS/Gurobi archive with 3,456 distinct configuration keys, 288 abstentions and 768 dyadic exports. Running the same script with `--root` set to `results-revision/research-audit/budget-final-cplex` checks the CPLEX arm: 1,728 keys, 144 abstentions and 384 dyadic exports. Thus the two invocations cover all three solvers: 5,184 planned configurations, 432 abstentions and 4,752 actual optimizer calls, with 1,152 exact dyadic export checks. Both use the same 96 models, whose CPLEX copies are byte-identical. `scripts/research/verify_stress_coverage.py` reconciles the per-root verification records; `make research-check-python` runs both invocations. Factor admissibility and dyadic multiplication are checked rationally. The separate `research-claims` audit also enumerates and checks the saved acceptance decisions for CPLEX.

A verified optimum requires an exactly feasible rounded binary point attaining the enumerated objective, a returned point within 10^{-9} of that rounded point, and bounds respected within 10^{-9}. Raw-point residual budgets and integrality errors are separate diagnostics. In particular, the verified-optimum label concerns the rounded assignment; it does not claim exact feasibility of the raw floating-point point. A failure of this conservative check does not by itself identify a solver bug.

Rational checks interpret each stored binary64 datum exactly. They do not recover unrecorded real-valued physical data. Arbitrary-factor multiplication may change the represented row by rounding even when its factor lies in the exact admissible interval. The dyadic routine avoids this by intersecting the coefficient/budget interval with exact exponent restrictions on every coefficient and the RHS. It compares candidate centering objectives as rational numbers, because floating logarithms can misclassify a near-midpoint as a tie.

Tests compare continuous interval projection with an independent linear program in log coordi-

nates, check positive re-emission, incompatible specifications, subnormal and overflow boundaries, and a cancellation example where a floating-point residual misses an exact unit violation. A native C++ integration test exercises the actual LP writer with very small coefficients, RHS, bounds and objective values. The serializer uses `max_digits10` and general decimal notation so binary64 values survive decimal round-trip parsing. Serialization correctness and exact multiplication are separate conditions; solver-internal transformations remain outside both guarantees.

S11 Activation contracts and exact feasible witnesses

This exploratory extension uses the six archived SG-Ter-Mer inputs without the documented total/local-demand inversion: instance001, instance005, instance022, instance027, instance031 and instance046. It does not repair or replace inputs. The existing numerical precision levels are applied explicitly to reservoir-volume activation thresholds. They are study objectives, not operator tolerances or measurements of error in the nonlinear plant model.

The coefficient-only L2 control projects its row factor onto $[10^{-9}, 10^{9}]$, scales the whole row and RHS, and uses no identity band. This is a separate control, not a renaming of the historical Flat-L2 policy. Budget-L2 uses the same kernel and adds the admissible residual interval. On a volume activation $M\varphi - v \le M - V$, the integer reserve is $E = M\gamma$. Default $\gamma = 10^{-5}$ exhausts the three volume budgets in this example. The guided allowance is the largest power of two not exceeding $\delta_V/(2\max_r M_r)$, capped at 10^{-5} and checked against Gurobi's supported minimum. At demanding precision it is 2^{-28} in all six models.

For activation rows with finite boxes $|x_j| \le X_j$, we also reserve

$$\zeta_r = 2^{-52}\left(\sum_j |a_{rj}|X_j + |b_r|\right).$$

Normal binary64 products and round-trip decimal serialization give a conservative forward-error allowance independent of the positive row factor. The selector uses $\delta_r - E_r - \zeta_r$; rational checks subsequently verify the actual exported defect against that allowance. Normality, coefficient limits and every factor interval are checked. The resulting demanding projection changes exactly the 1,440 activation rows in each model and no other L2 factor. On instance046, α increases from 10^{-9} to approximately 1.87×10^{-8}. The allowance does not assert that a solver enforces its nominal tolerance: observed exported and original residuals remain separate checks.

A rounding decision in original units. In the stored Base point of instance046, hour 18 has $\varphi = 0.9999996126403118$, $v = 2066844230.148119\,\mathrm{m}^3$ and a credited-power component $\theta = 13.60595418958399\,\mathrm{MW}$. The activation is $12494793414.450399\,\varphi - v \le 10427944344.323$; the associated power bound is $\theta \le 25.759667882689403\,\varphi$. For the stored binary64 data, fixing $\varphi = 1$ with continuous quantities unchanged violates the activation by exactly $2537543057/524288\,\mathrm{m}^3$. Fixing it to zero instead violates the power bound by 13.606 MW. Thus neither binary choice meets the demanding or reference pair of study budgets at that fixed continuous point. This certifies a need to change continuous quantities or reject that rounded interpretation, not infeasibility of the MILP.

Comparison protocol and endpoints. The principal comparison has 72 assignments: six models, three seeds (1, 7, 42) and four policies. Gurobi 13.0.2 uses four threads, 120 s, relative gap 1%, automatic internal scaling and presolve. L2-matched-int and Budget-L2 share $\gamma = 2^{-28}$ and

Table S11: Activation follow-up on six models and three seeds. All 18 calls per policy report completion. The Budgets column counts integer-rounded points passing exact original-unit checks on all audited physical rows and activations. $B > U$ counts reported bounds above an exact feasible objective of the same imported export, allowing the larger of 10^{-6} objective units and eight bound ulps for display rounding. Gap flags count cases satisfying $U < C - 0.01|C|$, incompatible with the requested completion criterion. Neither an unflagged bound nor budget compliance certifies global optimality. Work is the geometric mean over all assigned calls, including flagged completions; it must be interpreted with those flags.

Policy	Calls	Budgets	$B > U$	Gap flags	Work
GM	18	18	3	0	1.397
L2, matched γ	18	18	7	6	0.958
Budget-L2	18	18	1	0	1.045
L2, global tight	18	18	6	5	1.103

primal tolerance 10^{-6}; GM uses default integer tolerance 10^{-5}. L2-global-tight sets both tolerances to 10^{-9}. An eight-call, 60-second pilot on seed 17 is excluded. Additional controls on instance046 use the same three principal seeds (27 assignments). A subsequent presolve-off comparison fixes new seeds 11, 23 and 47 before its nine calls. Preset reference/permissive sensitivity adds six calls. There are 114 nonpilot calls, all retained, with their native logs, parameters and incumbent vectors. Calls are sequential and these timings are not pooled with earlier campaigns.

Exact witness construction. Observed exactly integral states are fixed, substituted into the rows, and the remaining LP is solved with SoPlex 8.1.0 and GMP 6.2.1 in rational mode. Its coefficients are full plain decimal expansions of the original binary64 numbers, not newly rounded decimal data. A separate Python `Fraction` checker verifies all 10,080 original rows, all 6,120 variable bounds and all 1,440 integer domains, and recomputes the exact objective. Twenty witnesses cover the native GM, L2-window and Budget-L2 exports in the six models, Base on instance046, and its reference-budget export. They are feasible points supplying upper bounds, not global optima. Each comparison identifies the actual imported model by SHA-256. The Base witness also applies to both dyadic controls after verifying exact proportionality of every coefficient and RHS. This prevents export-rounding differences from serving as an unsupported equivalence assumption. Tight-M/indicator controls have primal checks but no separate native-export witness in the bound comparison.

In the fresh-seed comparison, L2 gives a $4{,}837.338\,\mathrm{m}^3$ activation residual in all three calls. Budget-L2 reduces the maximum to $1.4782 \times 10^{-5}\,\mathrm{m}^3$. Global tightening reduces it to zero but retains the cost near -572330 and zero reported gap, contradicted by an exact feasible cost near -799086. The geometric mean optimizer work is 0.597 for L2, 4.485 for Budget-L2 and 0.577 for global tightening. The fast controls therefore cannot be regarded as equivalent successful computations.

Selection and round-trip export are measured separately in three fixed-order paired repeats per model, including budget-specific unit lookup and reserves but excluding common LP reading. Every timed export is byte-identical to its frozen counterpart. Median times are 0.354–0.360 s for L2-window and 0.497–0.505 s for Budget-L2, an additional approximately 0.14 s. The solver-call times include import and optimization, exclude these preparation measurements and verification, and are preserved separately. All data and solver-free replay programs are under `exploration/oms-followup-20261014`. The observed benefits concern the chosen target,

Table S12: Controls on instance046, with columns defined in Table S11. Upper panel: seeds 1, 7 and 42; presolve is automatic except where stated. Lower panel: fresh seeds 11, 23 and 47 with presolve disabled throughout. A dash indicates that no native-export witness was supplied for that reformulation, not that its bounds were certified. The dyadic L2 control still has one contradicted completion despite exact row proportionality. NumericFocus 3 is an effective existing alternative.

Policy	Calls	Budgets	$B > U$	Gap flags	Work
Base	3	0	0	0	51.646
Ruiz-R4	3	1	–	–	50.305
L2, presolve off	3	0	3	3	0.578
L2, NumericFocus 3	3	3	0	0	26.629
L2, dyadic	3	3	1	1	8.887
Budget-L2, dyadic	3	3	0	0	13.479
GM, IntegralityFocus	3	3	3	0	13.801
GM, tight M	3	2	–	–	12.594
GM, indicators	3	3	–	–	14.724

Policy	Calls	Budgets	$B > U$	Gap flags	Work
L2, matched γ	3	0	3	3	0.597
Budget-L2	3	3	2	0	4.485
L2, global tight	3	3	3	3	0.577

Table S13: Preset numerical volume targets on instance046 with seeds 1, 7 and 42. Changed rows counts differences from the coefficient-window L2 export. The demanding row is the corresponding subset of the principal comparison. All nine points meet their primal budgets, while the looser specifications permit contradicted completion claims. This is sensitivity of one configuration, not an estimated failure rate.

Scenario	δ_V (m^3)	γ	Changed rows	Budgets	Gap flags
Demanding	100	2^{-28}	1,440	3/3	0/3
Reference	1,000	2^{-25}	1,440	3/3	1/3
Permissive	10,000	2^{-22}	0	3/3	3/3

generator, solver and settings; the reference sensitivity and remaining bound discrepancies explicitly exclude an optimality guarantee.

S12 Sensitivity to incumbent initialization

Diagnostic question and protocol. This auxiliary experiment tests whether the observed completion advantage of external scaling persists when both representations receive the same incumbent. Three known SG-Ter-Mer models are selected retrospectively: instance001 and instance003 previously favored GM, while instance042 favored Base. They are not an independent validation sample. Each retains 6,120 variables, 10,080 rows and 1,440 binaries. The bank uses native MPS inputs and frozen power-of-two GM factors, rather than the historical LP exporter and arbitrary-factor policies; it does not retrospectively identify the cause of that campaign. Base and GM change only complete rows and their RHS, with exact proportionality checked. The GM policy uses the coefficient window $[10^{-9}, 10^{9}]$ and an identity band $[0.5, 2]$ on proposals, followed by projection onto the admissible power-of-two grid.

Table S14: Auxiliary common-incumbent diagnostic on three known cases and seeds 503, 509 and 521. Completion counts calls reaching the requested native gap, not certified global optimality. Work is the median over all three assigned calls, including censored ones; ratios involving those censored calls are not speedups. Both zero work values on warm instance042 are native reports, not zero total cost, and are not divided into ratios.

Case	Initialization	Base completes	Base work	GM completes	GM work
001	Cold	0/3	12.0001	3/3	5.4909
001	CommonStart	3/3	6.4158	3/3	6.7719
003	Cold	0/3	12.0002	3/3	4.3099
003	CommonStart	3/3	4.7903	3/3	4.6932
042	Cold	3/3	0.3284	3/3	0.4928
042	CommonStart	3/3	0.0000	3/3	0.0000

Base/GM, without a supplied incumbent (Cold) or with a common MIP start (CommonStart), and three new seeds (503, 509, 521) give 36 sequential MILP calls. Gurobi 13.0.2 uses four threads, limits of 12 seconds and 12 work units, relative gap 10^{-3}, primal tolerance 10^{-6}, integer tolerance 10^{-5}, IntegralityFocus=0, and automatic presolve and internal scaling. The schedule and source/input hashes are fixed before the calls. All assignments, limits, native logs, candidates and endpoints are retained.

Common reference and its cost. For each case, only the integer pattern of the archived GM run at seed 307 is used. A single SoPlex 8.1.0/GMP continuation, limited to 15 seconds, produces an exactly feasible point and matching primal/dual values for that fixed-pattern LP. These are not global MILP optima. The same binary64 conversion of the rational point is supplied to Base and GM; original-row, exported-row, bound and integer checks are retained separately. All 18 starts are loaded, and their integer pattern and objective are observed at node zero and zero recorded work. Acceptance is checked rather than assumed.

The three new reference acquisitions, including LP export and checking, take 1.79–1.84 seconds each. Their patterns were obtained by earlier GM runs requiring approximately 8.186, 4.931 and 0.451 native work units for 001, 003 and 042 respectively. The reference is therefore offline diagnostic information, not a free input to a proposed solution algorithm. Acquisition, construction and checking costs are separate from the optimizer-work comparison in Table S14.

With the common start, the paired GM/Base work ratios are 1.0555, 1.0519 and 1.3686 on 001, and 0.9957, 1.1114 and 0.7683 on 003. Thus the cold-start completion advantage disappears in both cases; GM has higher work in all three warm 001 pairs and mixed differences on 003. Where a root relaxation is logged, objectives coincide at printed precision in all 15 Base/GM pairs. This is neither exact LP certification nor an independent contribution: positive row scaling preserves the mathematical LP in exact arithmetic. Some root objectives change between Cold and CommonStart, and no root LP is logged for warm 042. A start can affect objective cutoffs, presolve and later search; the intervention does not isolate one heuristic or prove a unique cause of the effect.

Independent checks and reproducibility. All 15 unique final integer patterns enter a queue fixed before the posthoc exact-LP checks, with a five-second limit and no replacements. Thirteen have matching exact primal/dual optima; two remain unknown, both from Base-Cold 001, which did not reach the requested gap. The common-start comparisons have certified continuations at comparable costs. Conditional pattern optima are not global MILP lower bounds. No reported

bound exceeds the common exact feasible objective in this experiment; absence of contradiction does not validate global bounds.

The independent replay reconstructs references and fixed-pattern LPs from original binary64 rows, verifies their rational primals and duals, checks row proportionality, recomputes candidate objectives and final-point residuals after integer rounding, and verifies start acceptance and all final-pattern assignments. Four endpoint tests passed before the MILP calls. Data, protocols, sources, native records, exact certificates and the two unknown outcomes are supplied under `exploration/oms-root-attribution-20261015/`. This selected-case diagnostic qualifies mechanism attribution; it provides neither a general scaling ranking nor a new incumbent-generation algorithm.

S13 Matched IntegralityFocus control and complete workflow cost

Twelve principal calls repeat instance046: L2-matched-int/Budget-L2, IntegralityFocus=0/1 and seeds 11, 23, 47. Each fresh process constructs the original C++ model once and runs selection, export, import, solve and verification. Exports match their historical SHA-256 values. Gurobi 13.0.2 uses four threads, 120 seconds, 1% gap, primal/integer tolerances $10^{-6}/2^{-28}$, presolve off, automatic internal scaling and no MIP start. Budgets, reserves and factors are unchanged.

IF1 and IF0 produce identical saved points, objectives, bounds and work in all six matched pairs. All returned integer coordinates are exactly integral. L2 retains the 4,837.338 m^3 activation violation; Budget-L2 limits it to 1.4782×10^{-5} m^3, while two of its three reported bounds still exceed exact feasible objectives. Thus IF1 does not repair this configuration; the result makes no claim about its general effectiveness or MILP optimality.

The three Budget-L2 Work values, unchanged from the earlier presolve-off campaign, are 3.6054776613, 4.9754950464 and 5.0281587478 for seeds 11, 23 and 47. Their geometric mean is 4.4847271866 (4.485 in Table S12 and historical main-text Table 11 of v1.3.8); their median is 4.9754950464 (4.975 here and in the workflow note). The different summaries, rather than a configuration change, explain the apparent discrepancy.

The first twelve-call pass repeatedly copied binding vectors in the import checker. Caching them once and repeating all treatments leaves every point and native endpoint unchanged. These 24 calls are not pooled as independent samples; principal timings use the second pass. Compilation and prior witness acquisition are offline costs; revalidating both exact witnesses is timed separately. The independent replay checks 103,680 physical row/point combinations before/after rounding and the same-model bound/gap refutations. Sources, all stages, both passes, the two existing witnesses and four endpoint tests are supplied under `exploration/oms-if1-workflow-20261015/`.

Table S15: Matched IF control and full prototype costs. Upper: budget and bound/gap definitions follow Table S11; Work is the median. Lower: median wall seconds. Generate includes C++ launch, input, construction and Base writing; Select/export covers factors and scaled writing; Import/check covers native setup and model equality. Total includes the entire fresh process, also startup, reading, metadata, integrity, extraction and persistence. Stages shown are not exhaustive; medians need not add. Shared-machine timings describe this prototype and do not establish general performance.

Policy	Calls	Budgets	Max. v_V (m^3)	$B > U$	Gap flags	Work
L2-window, IF0	3	0/3	4,837.338	3	3	0.597
L2-window, IF1	3	0/3	4,837.338	3	3	0.597
Budget-L2, IF0	3	3/3	1.478×10^{-5}	2	0	4.975
Budget-L2, IF1	3	3/3	1.478×10^{-5}	2	0	4.975

Policy	Generate	Select/export	Import/check	Solve	Verify	Total
L2-window, IF0	0.115	0.428	0.114	0.639	0.269	2.070
L2-window, IF1	0.115	0.367	0.099	0.533	0.228	1.768
Budget-L2, IF0	0.065	0.514	0.097	4.788	0.237	6.231
Budget-L2, IF1	0.064	0.528	0.099	4.955	0.239	6.380

S14 Fully matched accuracy and numerical-remedy comparison

A fixed factorial comparison uses the same six demand-inversion-free models as Section S11, seeds 11, 23, 47, automatic/off presolve, and L2, Budget-L2, L2-tight and L2-NF3. The 144 terminal comparisons use Gurobi 13.0.2, four threads, 120 seconds, 1% gap, ScaleFlag=−1, IntegralityFocus=0 and no MIP start. L2 and Budget-L2 share primal/integer tolerances $10^{-6}/2^{-28}$; L2-tight uses 10^{-9} for both; L2-NF3 adds NumericFocus=3 to the matched L2 settings. Each fresh process constructs the C++ model once. Base and scaled exports reproduce their archived hashes exactly; only the 1,440 activation factors differ between L2 and Budget-L2 in each model. The original-unit targets, conversion, integer and export reserves are unchanged.

The randomized order and endpoints were fixed before optimization. A host reboot interrupted the 38th attempt; its partial files are preserved, the preceding 37 complete calls are retained, and only the interrupted assignment is repeated before continuing the original order. Thus 145 native attempts produce 144 terminal comparisons, with one separately recorded external failure of delivery. There are no new reference LP solves. The reader-only replay rechecks twelve supplied exact feasible witnesses and all saved candidates using rational binary64 arithmetic.

All policies meet the primal budgets under automatic presolve. With presolve off, only L2 on instance046 fails them, in all three seeds. Budget-L2 and both numerical remedies meet every primal budget. Budget-L2 retains one/three bound discrepancies under automatic/off presolve, whereas NumericFocus=3 has none detected in either panel. Their paired geometric-mean Work ratios, Budget-L2/NumericFocus=3, are 0.361 and 0.227 respectively over all 18 pairs per panel. These compare terminal primal delivery, not equally certified optimization outcomes; absence of a known contradiction is not a bound certificate. The fully matched controls clarify the cost and remaining limitation without extending primal recovery to a second model.

Complete process times include startup, C++ generation/Base writing, metadata, factor selection, scaled export, native import/equality checks, optimization, extraction, verification and persistence. Disjoint stages and objectives are retained for every call. Compilation and prior witness acquisition are offline; current witness revalidation is timed separately. Wall times span a reboot and changing shared-machine load, so native Work is the primary effort measure. The endpoint is terminal budget

Table S16: Fully matched numerical remedies. Upper: each policy has 18 terminal calls per presolve setting; "Budgets" counts points meeting all 4,320 audited rows after integer rounding. Max. v_V is the largest activation violation in m^3; $B > U$ and Gap flags follow Table S11. Limits counts native time limits. Work is the geometric mean of all terminal calls, including flagged completions; Total is the median entire fresh-process time in seconds. Lower: per-model compliance among three seeds with presolve disabled. Absence of a bound or gap flag does not certify optimality. The interrupted attempt is retained separately rather than imputed or counted as a completed replicate.

Presolve	Policy	Budgets	Max. v_V	$B > U$	Gap flags	Limits	Work	Total
Auto	L2	18/18	4.768e-07	5	3	0	1.59	1.944
Auto	Budget-L2	18/18	0.0001123	1	0	0	1.106	2.17
Auto	L2-tight	18/18	0	5	3	0	1.516	1.894
Auto	L2-NF3	18/18	0	0	0	0	3.063	2.295
Off	L2	15/18	4837	8	8	0	0.2904	1.343
Off	Budget-L2	18/18	1.478e-05	3	0	0	0.6023	1.795
Off	L2-tight	18/18	2.384e-07	8	8	0	0.2888	1.369
Off	L2-NF3	18/18	0	0	0	0	2.648	2.22

Instance	L2	Budget-L2	L2-tight	L2-NF3
instance001	3/3	3/3	3/3	3/3
instance005	3/3	3/3	3/3	3/3
instance022	3/3	3/3	3/3	3/3
instance027	3/3	3/3	3/3	3/3
instance031	3/3	3/3	3/3	3/3
instance046	0/3	3/3	3/3	3/3

delivery at the native stopping rule, not time to the first verified incumbent or a precision–cost frontier over multiple budgets. Known models and previously used seeds provide within-family confirmation, not external validation. Sources, every outcome, the interrupted attempt and paired cost summaries are supplied under `exploration/oms-matched-accuracy-20261016/`.

S15 Multi-budget precision–effort sensitivity

The six known SG-Ter-Mer models are paired at seeds 59,71,89 under automatic/off presolve, seven volume targets and a fixed 0.01 MW power target. The common integer tolerance is 10^{-9}, selected before optimization to study the full budget range; it differs from 2^{-28} in Table S16. Other settings match that study, including 120 seconds, four threads and a 1% requested gap. L2-tight uses both tolerances 10^{-9}; L2-NF3 adds NumericFocus=3 to the matched L2 control. Of 360 randomized assignments, 30 abstain before solving because the uniform integer-rounding reserve exhausts $10\,m^3$ in five models. The sixth admits that budget. This is incompatibility of a sufficient contract, not MILP infeasibility. All remaining 330 calls and 31 exact native-export feasible witness acquisitions are retained. No fresh MILP run is needed for the replay.

The criterion fixed before solving requires at least four models to meet the targets in all three seeds at two adjacent budgets, with lower paired geometric work than NF3, no more detected bound contradictions and at most 1% worsening in candidate cost. All six qualify at 30 and $100\,m^3$

Table S17: Prespecified sensitivity. Calls counts executed Budget-L2 calls out of 18 assignments; the missing 15 at $10\,\mathrm{m}^3$ in each panel are abstentions. Budgets counts successful rounded points. B/G denotes bound/gap contradictions against exactly feasible witnesses of the same binary64 export. NF3 has zero detected B/G flags and satisfies all targets in both panels. Work and complete-time ratios are paired geometric means against NF3 on the same executed cases/seeds. All controls are resolved once per case/seed/presolve and checked at every target. At 3,000 and 10,000 m^3, Budget exports equal L2; these are not independent representations. Lower effort or absence of a detected contradiction does not certify MILP optimality.

Presolve	δ_V	Calls	Budgets	B/G	L2 B/G	W/W_{NF3}	T/T_{NF3}
Auto	10	3/18	3/3	0/0	5/4	0.567	0.969
Auto	30	18/18	18/18	0/0	5/4	0.411	0.704
Auto	100	18/18	18/18	0/0	5/4	0.362	0.629
Auto	300	18/18	18/18	1/0	5/4	0.364	0.688
Auto	1000	18/18	18/18	6/6	5/4	0.496	0.874
Auto	3000	18/18	18/18	5/4	5/4	0.487	0.811
Auto	10000	18/18	18/18	5/4	5/4	0.487	0.854
Off	10	3/18	3/3	0/0	9/9	0.599	0.943
Off	30	18/18	18/18	1/0	9/9	0.226	0.579
Off	100	18/18	18/18	3/2	9/9	0.188	0.561
Off	300	18/18	18/18	6/6	9/9	0.119	0.499
Off	1000	18/18	15/18	9/9	9/9	0.115	0.500
Off	3000	18/18	15/18	9/9	9/9	0.114	0.479
Off	10000	18/18	18/18	9/9	9/9	0.114	0.496

under automatic presolve. Both policies pass every primal target and have zero detected bound/gap contradictions there. Work ratios by model range from 0.128 to 0.668 at $30\,\mathrm{m}^3$, and 0.072 to 0.585 at $100\,\mathrm{m}^3$. This is a descriptive comparison within known models, not external validation or certification of their reported MILP bounds. Primal recovery over L2 still occurs only on instance046 with presolve disabled.

Complete process time includes startup, generation once, metadata, selection, export, import/equality checking, solving, extraction, original-row verification and persistence. Compilation and witness acquisition are separate. On the shared host, the aggregate time ratios at $30/100\,\mathrm{m}^3$ are 0.704/0.629; the $30\,\mathrm{m}^3$ cycle is nevertheless slower in three inexpensive models. No isolated hardware speedup is claimed. The 36 automatic-presolve Budget points in this band satisfy the observed exported-row hypothesis: no audited row exceeds $\epsilon = 10^{-6}$ and the maximum is below 1.5×10^{-10}. That observation is checked separately from factor admissibility and original-unit residuals. With larger budgets or presolve off, Table S17 retains adverse results. Sources and all outcomes are in `exploration/oms-frontier-20261016/`.

Table S18: Two separate Budget-L2 implementation benchmarks. Preparation reports medians of seven repeats per arm/model; speedup is Python/C++. The complete cycle reports geometric means over seeds 59, 71, 89; its ratio is C++/Python, with aggregate paired ratio 0.824. The former includes startup, generation, selection, export and persistence; the latter additionally includes native import/data checks, solving and physical verification. Compilation and prior witness acquisition are separate. These are shared-host measurements, not general hardware speedups.

Standalone preparation

Model	Python (s)	C++ (s)	Preparation speedup
001	1.318	0.265	4.97
005	1.167	0.315	3.70
022	1.167	0.315	3.71
027	1.368	0.215	6.37
031	1.318	0.315	4.18
046	1.168	0.315	3.70

Complete process

Model	Python (s)	C++ (s)	C++/Python time
001	4.378	3.974	0.908
005	1.636	1.218	0.744
022	1.620	1.184	0.731
027	2.874	2.472	0.860
031	1.586	1.185	0.747
046	10.555	10.416	0.987
All 18 pairs	–	–	0.824

S16 Isolated native-selector integration

An experimental hook applies the selector to the C++ generator's in-memory rows, choosing all factors before modifying any row and exporting only once. The original generator and its formulation are unchanged. Both arms use the same isolated $-O2$ executable: Python disables the hook, exports Base, reconstructs metadata and applies the reference selector; C++ enables the hook. Exact rational intervals and reserves use Boost 1.74 multiprecision. At 0.01 MW/100 m^3 and integer allowance 10^{-9}, all factors, matrix/RHS values, objectives, bounds and integer domains match the Python implementation on all six models, for L2 and Budget-L2. The independent replay checks 120,960 paired rows and 25,920 contracts.

The corrected pass contains 168 timed preparations across both kernels, plus 24 preflight preparations. An earlier complete pass is retained: its C++ audit CSV failed to quote row names containing commas. After correcting that format, the entire identical design was repeated; every model export matches the first pass. Thus 336 timed and 48 preflight preparations are supplied, with no optimizer call. Part of standalone startup savings may overlap imports needed by a full solver workflow, so the speedup is not applied to historical optimization times. The prototype is under `exploration/oms-native-selector-20261016/`.

The subsequent complete-cycle benchmark fixes 36 calls in randomized paired blocks on the same six models. Both arms read original-model metadata within the measured process; Python disables the native selector and C++ enables it in the same executable. All ordered binary64 model data, including row and column order, match. All 18 pairs have identical points, native endpoints and work. Paired time ratios by model range from 0.731 to 0.987, with aggregate 0.824; the 3.70–6.37

preparation speedups therefore are not extrapolated. No other research solver campaign was run concurrently with this benchmark; system load was recorded and all repetitions retained. Complete records and independent checks are under `exploration/oms-followup-20261017/native-cycle/`.

S17 Exact global-LP reference intervals

The six original Base models admit a supplementary verification independent of their reported solver bounds. All 1,440 binary variables are relaxed, preserving their $[0, 1]$ bounds; no integer pattern is fixed in this LP. SoPlex 8.1.0 with GMP obtains an exact primal/dual solution using the rational values of the original binary64 coefficients, RHS, costs and bounds. The checker independently rebuilds all inequalities, including variable bounds, verifies dual signs and $A^T\pi = c$, and checks primal/dual objective equality. The dual therefore provides a genuine global lower bound L for the original MILP.

Separately, exactly feasible integer witnesses provide U. Their construction uses a fixed-pattern LP, but only their verified primal vectors are used here; that LP's dual is not a global MILP bound. Table S19 reports the resulting intervals. Negative objectives use width $(U - L)/|U|$. Equality is tested rationally, not by comparing rounded displays. Three reference witnesses are certified within 1%, and instance031 is certified exactly optimal. Wider intervals on the other three models do not certify their reported 1% gaps. These are original-Base reference certificates, not a claim that a floating row export, a tolerance-feasible native vector or every point in Tables S16–S17 has a certified optimality gap.

The six global-LP construction/solve/check cycles cost 23.98 seconds in total on the shared host; acquisition of the six integer witnesses is additional and recorded separately. Replaying supplied primal/dual certificates requires no optimizer. Sources and duals are under `exploration/oms-global-lp-20261016/`; the supplied `exploration/oms-joint-contracts-20261016/base-witnesses/` holds the original-Base integer witnesses. The separate row/column research prototype is outside the row-scaling claims of this manuscript.

Table S19: Verified original-model intervals from six unfixed exact LP relaxations and six complete exact integer witnesses. Displayed costs are rounded; certificates retain exact rationals. A width over 1% is an unresolved interval, not evidence that the integer optimum is known or that the relaxation gap equals that width.

Model	Global LP L	Integer witness U	Width (%)	Verified conclusion
001	-1208494.856	-989879.474	22.085	Interval only
005	-2558473.761	-2540302.931	0.715	Within 1%
022	-293986.477	-292995.512	0.338	Within 1%
027	-1615641.968	-1598079.853	1.099	Interval only
031	-1542528.000	-1542528.000	0.000	Exact optimum
046	-840709.945	-799086.098	5.209	Interval only

S18 Independent batch-scheduling transfer

Eight prescribed horizons (8,10,12,16,20,24,32,40) use the archived STN-Scheduler implementation of the Kondili batch-scheduling example, independent of the hydrothermal generator. Horizons 10 and 12 are known anchors from an earlier pilot that found no added primal benefit; the other six are new horizons of the same external example, not six independent application families. Source commit, license and complete model data are retained in the artifact. Mass budgets are 10^{-5} kg (10 mg), logical budgets 10^{-5} and financial budgets 10^{-3} source monetary units. These are numerical study goals, not industrial accuracy standards.

All four controls use exact dyadic row/RHS export, the same coefficient window, integer tolerance 10^{-9}, automatic presolve, four threads, a 30-second limit and a requested 1% gap. Seeds 131,137,149 and all 96 assignments are fixed before solving. L2-tight changes FeasibilityTol from 10^{-6} to 10^{-9}; NF3 adds NumericFocus=3. The integer tolerance is already 10^{-9} in every arm. A pre-optimization construction check exposed the upstream dependence of ConstraintList names/order on Python sets. The correction verifies the complete mathematical row multiset, including duplicates, then restores the already frozen row order. The failed preflight and source-only amendment are retained; no model, export, assignment or optimizer result was changed.

Table S20 verifies transfer of the procedure: Budget-L2, L2 and L2-tight each pass 24/24 checks, and NF3 passes 23/24. At horizon 40 and seed 149, the NF3 point has $W = 0$ and $B = 1.110407442 \times 10^{-4}$ kg in the capacity row $B - 200W \leq 0$. Its violation is 111.041 mg against the 10 mg target, despite exact integer coordinates. L2's factor 2^{-8} yields observed exported violation $4.338 \times 10^{-7} < 10^{-6}$. Budget-L2 selects 2^{-3}: its conditional rounded-original-row bound is $8 + 0.2 = 8.2$ mg, within the target. The checked Budget-L2 point meets the target. Matched L2 also passes all checks; this example does not establish a unique primal recovery over that control or general computational superiority. Native solves are short and process overhead dominates.

Eight separate NF3 reference calls at seed 127 supply patterns for eight single 10-second exact LP attempts. One yields an independently verified feasible witness; seven terminate without a certified solution and are retained as unknown, not infeasibility proofs. Thus only 12 comparison-bound checks have this reference; the other 84 are unknown. No replacement pattern was sought. These references do not certify the native reports. The independent replay checks all 104 saved points, factor intervals, exact proportionality and the retained witness without optimizing. Data and

Table S20: Independent batch-scheduling checks. Entries count returned points meeting all original-row and variable-bound budgets after integer rounding. All integers are retained. Changed rows compares exact dyadic Budget-L2 with matched dyadic L2. The final row totals 24 calls per policy. Budget delivery is not exact feasibility or a certificate of the reported MILP bound or gap. NF3's sole failing point has 111.041 mg capacity violation against 10 mg; the contractual bound is 8.2 mg.

Horizon	Changed rows	L2	Budget-L2	L2-tight	L2-NF3
8	72	3/3	3/3	3/3	3/3
10	88	3/3	3/3	3/3	3/3
12	104	3/3	3/3	3/3	3/3
16	136	3/3	3/3	3/3	3/3
20	168	3/3	3/3	3/3	3/3
24	200	3/3	3/3	3/3	3/3
32	264	3/3	3/3	3/3	3/3
40	328	3/3	3/3	3/3	2/3
All	1360	24/24	24/24	24/24	23/24

programs are under `exploration/oms-followup-20261017/external/`.

S19 Integer-tolerance sensitivity and activation-factor path

The follow-up matches six known dispatch models, seeds 59,71,89, automatic presolve, the 0.01 MW target and volume budgets 30/100 m^3 at integer tolerances 10^{-9} and 2^{-28}. All other native parameters match. Contract factors are recomputed consistently with the integer allowance, so this tests the whole workflow's tolerance sensitivity. The NF3 representation is unchanged and isolates its native parameter change. Fifty-four low-tolerance calls are reused from Table S17; the high-tolerance design adds 36 calls and 18 contractual abstentions. Twelve existing exact-export witnesses suffice; their equality to the corresponding new exports is checked and no LP is rerun.

Table S21: Matched tolerance sensitivity. Calls and abstentions refer to Budget-L2; Budgets counts successful rounded points. B/G uses the exact feasible references and thresholds of Table S16. NF3 has no detected B/G flags in either tolerance panel. Work ratios are paired geometric means against NF3 at the same tolerance. An abstention concerns the sufficient precision specification, not MILP infeasibility.

IntFeasTol	δ_V	Calls	Abstentions	Budgets	B/G	W/W_{NF3}
10^{-9}	30	18/18	0	18/18	0/0	0.411
10^{-9}	100	18/18	0	18/18	0/0	0.362
2^{-28}	30	0/18	18	–	–	–
2^{-28}	100	18/18	0	18/18	1/0	0.356

At 2^{-28}, the integer-rounding reserve alone exceeds 30 m^3 in every model. At 100 m^3, every point passes, with work ratio 0.356 but one contradicted bound. The zero-detected-contradiction result at the smaller tolerance therefore does not establish robustness across these values.

A separate replay uses existing exports only. At every tested volume target from 30 through 1,000 m^3, exactly 8,640 factors differ from L2, all on activation rows; 319,680 nonactivation rows across compatible exports are checked unchanged. With R_i denoting the recorded combined integer/export reserve, the observed activation factors satisfy

$$\alpha_i = \mathrm{ceil}_{64}\left(\max\left\{10^{-9}, \frac{\epsilon}{\delta_V - R_i}\right\}\right).$$

Independent arithmetic checks 53,280 compatible factors and 7,200 incompatible activation intervals directly, without importing the selector or experiment driver. The L2-admissibility thresholds $R_i + \epsilon/10^{-9}$ range from 1,008.779 to 1,013.068 m^3. Thus the factor magnitudes, rather than the number of modified rows, change within the favorable and adverse tested targets. All automatic-presolve outputs in that sensitivity meet the observed exported-row hypothesis, including cases with contradicted bounds. This locates the changed representation; it does not distinguish internal presolve, factorization or search causes of the observed bound discrepancies. The figure and complete row records are retained in `exploration/oms-followup-20261017/mechanism/`.

S20 Secondary audit of robustness and external costs

This analysis uses already observed outcomes from Tables S17 and S20. Its aggregation protocol was recorded before these calculations, after the campaigns had been inspected; it is a secondary descriptive analysis, not a new confirmatory experiment. All seven budgets, both presolve settings, every seed, abstention and adverse outcome remain in the accompanying CSVs. Table S22 gives the two favorable automatic-presolve targets and the complete external control comparison.

Table S22: Secondary checks on observed outcomes. Top: paired Budget-L2/NF3 work ratios on the six known dispatch models and seeds 59,71,89. Deletion columns give the range of geometric-mean work ratios after omitting each complete model or each complete seed in turn, retaining the remaining pairs. These ranges are not confidence intervals. Work/time wins count individual pairs with ratio below one; all 18 Budget-L2 points at each target meet the budgets without detected B/G contradictions. Bottom: all eight external batch-scheduling horizons and seeds 131,137,149, with paired work and complete-time ratios against L2. Unknown bound checks are retained: each policy has only three of 24 checks with an exact feasible reference, and 21 unknown. Time is measured on a shared host and excludes compilation and prior witness acquisition.

δ_V	Work wins	W/W_{NF3}	Omit one model	Omit one seed	Time wins
30	18/18	0.411	0.373–0.519	0.380–0.446	11/18
100	18/18	0.362	0.329–0.499	0.347–0.387	14/18

Policy	Budgets	W/W_{L2}	T/T_{L2}	Known bound checks
L2	24/24	1.000	1.000	3/24
Budget-L2	24/24	0.960	1.054	3/24
L2-tight	24/24	1.003	0.990	3/24
L2-NF3	23/24	1.732	1.004	3/24

All 36 dispatch work comparisons favor Budget-L2 at the tested 30 and 100 m^3 targets. Omitting any model or seed retains the aggregate work advantage. The result is therefore not driven by one observed model or seed; it remains restricted to known models and settings. Time improves in 11/18 pairs at 30 m^3 and 14/18 at 100 m^3. Maximum compared candidate-cost deterioration is 0.799% and 0.0074%, respectively; these are candidate comparisons, not certified MILP gaps. The integer-tolerance counterexample of Table S21 and adverse cells of Table S17 still apply.

External Budget-L2 has work ratio 0.960 and complete-time ratio 1.054 against L2; against NF3 the ratios are 0.554 and 1.049. It meets the numerical budgets without a complete-time advantage. Restricting the descriptive comparison to the six nonanchor horizons retains all 18 Budget-L2 and L2 primal passes; those are new horizons of the same external example, not new application families. The audit rechecks 222 dispatch pairs against saved raw outcomes and parent process timings, and checks geometric means independently using 80-digit decimal products. Scripts, all deletion results and the input hashes are supplied in `exploration/oms-evidence-audit-20261017/`. No optimization calls are added.

S21 Directed unscaled control of the favorable dispatch result

The original seven-budget study omitted Base at its matched settings. This subsequent directed study fixes its design after observing the earlier results, before its 36 Base/Base-tight calls. It uses the same six models, seeds 59,71,89, automatic presolve, four threads, 120 seconds, 1% gap, ScaleFlag=−1, IntegralityFocus=0 and IntFeasTol=10^{-9}, on the same shared host with Gurobi 13.0.2. Base means no external scaling; internal scaling remains automatic. Base-tight changes only FeasibilityTol from 10^{-6} to 10^{-9}. No starts, WorkLimit or generator changes are introduced. Each generated Base matches its historical hash; native matrix/RHS, objective, bounds and domains are preserved. The controls are solved once per model/seed and checked at both targets.

Independent replay checks the 36 new points and 36 saved Budget-L2 points against all 10,080 original rows before/after integer rounding, bounds, domains and objectives. It rechecks 18 existing

Table S23: Directed comparison: B is Budget-L2 and C is the named control. Both pass counts pairs meeting every rounded target at 0.01 MW and the listed volume budget, in m^3. Ratios are geometric means of 18 pairs. Max cost is the maximum $100(C_B - C_C)/|C_C|$ of recomputed candidate costs, not certified gaps. Initial work uses 36 new controls and 36 saved Budget-L2 calls; their times are not combined. The interleaved panel uses contemporaneous fresh-process repeats. All logs and points are retained.

Stage	δ_V	Control	Both pass	W_B/W_C	T_B/T_C	Max cost (%)
Initial	30	Base	18/18	0.360	—	0.8405
Initial	30	Base-tight	18/18	0.300	—	0.8320
Initial	100	Base	18/18	0.317	—	0.0074
Initial	100	Base-tight	18/18	0.264	—	0.0961
Interleaved	100	Base	18/18	0.317	0.621	0.0074
Interleaved	100	Base-tight	18/18	0.264	0.570	0.0961

rational feasible witnesses of the actual exports, without new reference LPs. All policies meet every physical target, with zero detected bound/completion contradictions or time limits. All stored integers are exactly integral. Maximum rounded nonphysical residuals are below 2.9×10^{-7} (Base), 5.4×10^{-13} (Base-tight) and 6.5×10^{-14} (Budget-L2); bound violations stay below 4.6×10^{-13}. Floating points are not exact MILP-feasibility certificates.

At $100\,m^3$, model-level work ratios range from 0.0422 to 0.9926 against Base and 0.0346 to 0.8042 against Base-tight. The advantage over Base on instance031 is small. Budget-L2 uses less work in 17/18 individual Base pairs and 18/18 Base-tight pairs, with cost worsening below 0.008% and 0.097%, respectively. At $30\,m^3$, worsening stays below 0.841% against either control. Both targets pass the locally fixed confirmation gate: lower work in aggregate and at least four models against both controls, all budgets met, no additional detected contradictions and at most 1% cost worsening. The conditional confirmation fixes 54 fresh-process calls at $100\,m^3$, in 18 consecutive model/seed blocks with randomized policy order. All 54 points, objectives, bounds, work and native endpoints repeat exactly; all budgets pass, with zero detected contradictions or time limits. Complete-process time includes startup, generation once, any selection/export, import/equality checks, solving, extraction, verification and persistence; compilation and historical witness acquisition are separate. Paired time ratios are 0.621/0.570 against Base/Base-tight. Only 9/18 pairs are faster against each: models 005,022,031 are slower. These are interleaved shared-host measurements, not an isolated hardware benchmark or validation on unknown seeds.

This fills the omitted baseline without establishing external generality, primal recovery over Base or a solver-internal cause. Base's water residuals are smaller, though both policies are far below budget. Absence of a detected contradiction does not certify global optimality. Designs, hashes, all outcomes and replay commands are in `exploration/oms-base-control-20261018/`; `make research-oms-check` checks them without optimizing. Its absent-history flag skips only unavailable frozen historical PDF copies, never scientific inputs or numerical checks.

S22 Reproduction and artifact layout

The accompanying locally frozen snapshot is `v1.3.15-oms-submission`, revised 18 October 2026 and supplied as Supplementary Material 2. Its metadata distinguish the publicly archived `v1.2-coap-submission` baseline from the new Ruiz comparison, complementary implementation check, expanded physical-budget audit, activation follow-up with exact native-export feasible witnesses and the auxiliary common-incumbent diagnostic, matched IntegralityFocus control, fully matched accuracy/remedy comparison, complete workflow costs, the multi-budget sensitivity, native-selector preparation benchmark, global-LP reference certificates and the directed unscaled-control study of the favorable dispatch result. `paper/submission/oms/OMS_DATA_MANIFEST.json` inventories every new experiment file and required replay program/input with a SHA-256; `SNAPSHOT.json` identifies the version and counts. All compressed new LPs, named bases, row factors, native logs, saved incumbents and physical-row audit inputs are supplied. The new-result replay uses these files without the historical bulk archive or any optimizer call, including the twenty supplied activation witnesses. SoPlex is only needed to generate new witnesses, not to check those provided. No separate Zenodo version DOI is claimed for the OMS snapshot.

The root README specifies the environment. These commands need no optimizer license:

```
make research-oms-check # includes matched controls and full workflow replay
make research-check-python # tests, saved binary audit, matrix hashes
make research-check # also builds/tests the native C++ exporter
make research-analysis # current tables and vector figures
make research-fixed-basis-check # rebuild all 1,530 bases from LP text
make paper # article and this supplement
```

The four audit/analysis commands above are self-contained in Material 2: they use the supplied new OMS files and stored historical analysis inputs. The stopped-campaign screening tables use the supplied `runs.csv` and `contradictions.csv` under `results-revision/solver-robustness/analysis/`; these are numerical screening records, not exact feasibility certificates. Historical basis reconstruction and rerunning its stress-sample precision calculation additionally require `fixed-basis-exports.tar.zst`, archived with `v1.2-coap-submission` at https://doi.org/10.5281/zenodo.23145294. That separate archive contains 337 MB compressed and 5.18 GB uncompressed in 11,585 files. `EXCLUDED.txt` identifies omitted bulk files; their sizes and hashes are in `paper/submission/data-deposit/MANIFEST.json` and `paper/submission/data-deposit/SHA256SUMS`. Extract at the repository root before `make research-fixed-basis-check`; the 1,530-basis reconstruction is not part of the self-contained four-command replay. The full export check needs GNU Make and a C++17 compiler. PDF compilation uses pdfLaTeX, BibTeX and standard TeX packages. Figures are generated with matplotlib from stored data or explicit mathematical expressions.

Fresh experiments write to separate directories and are not required for verification:

```
make research-grid GRID_OUT=results-revision/research-audit/reruns/grid
make research-fixed-basis \
  FIXED_BASIS_OUT=results-revision/research-audit/reruns/fixed-basis
make research-budget BUDGET_SOLVERS=highs \
  BUDGET_OUT=results-revision/research-audit/reruns/highs
```

The default two-solver budget rerun also requires Gurobi and a suitable license. Runtime depends on hardware and solver versions. Supplied snapshots and hashes identify the recorded runs; subsequent arithmetic repairs and comparisons against saved factors are documented separately. The accompanying source/data archive (Supplementary Material 2) preserves the inputs needed to regenerate the current main tables and figures.